

"Physics as Information Processing" ~ Ander Aguirre ~ Discussion 4

Source: [YouTube](#) Playlist: Physics as Information Processing ~ Chris Fields ~ 2023
Duration: 1:07:15 | Uploaded: Unknown Transcript method: auto_caption

foreign hello and welcome everyone it is Saturday August 26 2023 and we're here in physics as information processing in the fourth discussion thank you everybody who's joining live or watching live or watching and replay this should be a really fun discussion as usual so we're going to have under provide some slides and some initial clarifying questions for Chris and then we're gonna open it up for our guests to ask some questions and check in a live chat see what people are talking about so Andre to you for some slides and overview okay thank you Daniel so um last time we were starting to talk about multi-agent communication and log protocols and all of that so let's quickly go through go through the slides and uh sort of try to recall the main Concepts so um I guess from just to go back uh to session three for a second uh one takeaway was that when you have non-comedian sectors in a boundary that's necessarily going to require classical communication between them so that is more or less the whole point of today uh how non-commune sectors May communicate have to communicate classically um another uh introduction in in this last lecture was uh this new graphical notation uh tqft if you wish to call it that way for any of these clock ticks okay so so let's recall the the whole process of uh measuring a sector environment and writing it into a memory environment it can be simplified to this sort of tick the connect uh connects to two two portions of the boundary and you know uh everything we talk about log protocols will be will be written in this graphical notation let's keep going a little bit uh it's on the paper on uh sequential measurements from 20 2022 uh these are basically the formal mapping between qrf and uh topological Quantum field theories okay so we're more used used to the picture here on the left hand side of this slide right with with the qrf sectors but you know rhetorically speaking this is at least how they map okay and then uh from then uh we started talking about entanglement and classical Communications using this notation and and this is probably where my questions to Chris will will start uh so in section three uh we saw this idea of the fbp asymptotically being the principle of unitarity AKA driving the system the join system Alice involved uh to

entanglement which in the graphical notation amounts to them sort of uh completing a a cycle here where the QRS are perfectly aligned so that the S sector of one becomes the memory sector of the other and vice versa now I actually at this point already I have a question uh for for Chris I mean I have it right now here so yes two things here uh Chris number one uh is in in the in the log protocol picture we do have a distinguishing between a Quantum Channel and a classical Channel and if I understood that part correctly we said that passage of time or expenditure of energy free energy only corresponds to the classical tick so in this you know quantum entanglement picture where you don't quite have that is it fair to say that there's no such thing as as a classical take no passage of time no expenditure of free energy is this a degenerate picture of the log protocol where the sectors were aligned the right way and boom you got rid of that classical thing and and that's it and secondly I guess it's what does it mean for these two parts quote unquote uh in the from the paper to implement a joint uh unitary process you join Quantum process so yeah this is this is not a lock protocol clearly because uh there there is no classical uh communication and and in fact uh Alice is not divided into uh multiple compartments here uh Alice is just implementing one this this single Quantum process Q and uh because these sectors are are because these These qrf's are perfectly aligned we can think of of a PR remember uh probably the second session the the the picture that you were very interested in under how of the wick rotation uh reversing time between Alice and Bob so Ellis's clock in fact uh ticks in the opposite direction from Bob's clock because both of their internal clocks are counting incoming bits not outgoing bits so in this in the right side of this picture there there is no overall time you can think of Alice's clock uh ticking forward one unit and then Bob's clock ticking backward one unit so that uh the the information flow is completely time symmetric between Alice and Bob so we can think of of both of these processes q and Q Prime as bi-directional in time that's that's what it means to have a pure unitary process so no time a symmetry that's what it is right so there is no time asymmetry in this picture and once when you introduce classical communication of course you have to have a Time asymmetry right because uh you have this I mean classicality is defined by this notion of irreversibility which is just the same as As Time asymmetry and it's the irreversibility that burns energy so the there has to be energy flow between Alice and Bob to to drive what is in from a global point of view and only an apparent asymmetry but from the point of view of Alice 1 and Alice 2 as components of Alice oh they're the ones who see the so I guess uh just to tie back to the question of the weak rotation once this joint uh process is implemented I mean and iFly one can say is this cycle going clockwise or counterclockwise and it's is that

going to Define on originally whenever these guys ended up being lined up where the weak sign was coming from of yes whenever whenever Alice sees information flowing in Bob sees information flowing out so there their their Wick rotations they're going out opposite directions yeah uh very good and uh yeah the other the other thing to to think of in this picture which is it's impossible to draw all of these simultaneously is that if you if you think of the fep as fully aligning Alice and Bob then not only do these processes perfectly couple but the the area that's green uh in this right hand side kind of expands to fill the whole boundary so that in the in the complete limit of uh Alice being able to perfectly predict Bob and and vice versa then the whole boundary is green right so the oh there's no this distinction between the the S sector and the Y sector disappears yes yes that makes sense actually and just to conclude with this slide um back in the session too we saw something about uh agents requiring a free choice of reference frame for their qrf's which amounts to being able to choose how to measure a spin right right so it's part of this entanglement picture like do they do they just have to align the sectors or also the way they are measuring a spin on those sectors yes they they they have to align the way they're measuring the spin in those sectors that it's that lack of remember the the point that free choice of of local reference frame is required for separability so as the systems become fully entangled then their local reference frames for each qubit I'll become the same that might become necessarily the same thank you okay and then uh let's continue a little bit um so in in this in this talk we're mainly interested uh krfs that do not commute uh now maybe uh you can of course you can tie this back to what you were saying earlier about if you're perfectly entangled then the the sector is going to grow does it have anything to do with this so if q_1 and Q_2 here are do commute then how does how do you reduce this uh to the picture on the right hand side at the first class it looks like the picture on the left hand side encompasses uh more sectors of the boundary yeah the the point here is just that if if we go back and uh go back to the notation where I'll what we're we're representing by this this simplified notation is a collection of uh cone cocone diagrams and if the if the if these reference frames commute then one can take these two diagrams and and find a uh think of it in the information inflow Direction uh there will always be a common uh Co cone over the two sets of input classifiers if the two qrf's commute so what that means is that I go from a diagram with two sets of classifiers each going up to some Apex to a diagram with the merge set of classifiers going up to some some farther up the hierarchy Apex because they commute I can find that that sort of yet more abstract cocone that covers all of the classifiers in both qrf's so that just collapses of these pairs of green segments down to one green segment and I've drawn it the same size I

mean it would it would make more sense graphically I suppose to draw it right each one is twice as big or as each one is against the area but yeah keeping to this kind of very topological representation where the area isn't actually meaningful um it is convenient to draw it this way right the fact that the commute means that you will have that higher Apex in a way so you can bypass that that first nesting right but it comes to the number of degrees of freedom the boundary uh they're still the same and in terms of their yeah right okay yeah we're not we're not getting rid of any of the the degrees of freedom of the boundary that these uh curves are looking at uh yeah very good so the next slide um I found a very interesting one so these are no growth theorems things that Alex and Bob cannot do and they may you know it's mainly having to do our measuring their own entangled I meant entropy and Counting degrees of freedom I guess uh this has uh a very holographic feel to it and also maybe even a bit of a godo theorem feel to it so I don't know if Chris wants to comment on that well I'll just say that I I think this does have a kind of girdle undecidability feel to it and that there there are a large number of questions that Alice can't answer which uh certainly suggests a formulation in terms of incompleteness and this is the connection to girdle's theorem is actually something that Jim glazebrook and I are currently working on um and hopefully we'll be able to make that connection more precise but the the point of these statements is just to emphasize that a single Observer that's uncompartmentalized cannot uh determine anything about the entanglement entropy of the environment and this is this is just the result that to to measure entanglement you have to have you have to employ a lot protocols so you have to have classical communication this raises of a fundamental issue in the theory and that that uh classical communication is an exchange of information that cannot be represented as a single Quantum process you know hence you have language like the collapse of the wave function or these various other uh you know deed coherence etc etc these various ways of trying to capture this idea that something different and irreversible is happening as soon as you talk about classical the point is for for Alice to measure anything about the entanglement entropy of Bob she has to be sectorized so her entanglement entropy has to decrease but she can't know that she can't actually know that she has sector compartments when we think of Alice as an entire system from now on we're going to be talking about the very limited points of view of particular Observers and to to talk about that I think it's it's very important to keep these various limitations in mind because there's a real distinction between what we can say uh effectively from this kind of God's eye point of view that we're adopting as theorists where we say uh Alice has sectors that are communicating uh what Alice A system can do from her point of

view or uh her sectors can do from their points of view this is a distinction that's often ignored but it's important to keep it in mind awesome thanks for this lion enter yeah so before we continue so then uh I guess we get to the the proper material of of the last lecture which is what happens when you have that are not commutative such as two portions of dollars involvement that have to communicate classically this this is what requires the classical communication tick let me see uh so would you start first to say Chris uh this only such a classical Channel such as the one shown in this uh slide requires free energy is that right because there's some writing into a memory involved in here compared to the earlier slide where we had that closed loop in the entangle there was no expenditure of energy is that correct right so so here the the the example that came to mind here is uh exactly the process that we're that we're executing right now Alice has two components um and and we are currently in the position of those components and we're communicating classically with each other by uh employing a channel that's part of our shared environment so we're acting irreversibly on our shared environment it's that irreversible action on the environment that has the free energy cost for each of us we actually have to make noise to communicate you know our our laptops are actually consuming power all of this is classical I mean it's this irreversible irreversibility that defines classicality right and uh to continue with this question I guess uh it'll come in short in in later slides but the other side of these lock protocols is the Quantum Channel and my question to you uh Chris is just like there is an expenditure of time and energy involved in this classical Channel this is the Quantum Channel basically the opposite of that meaning it's it's it's on it's action is instantaneous entanglement presumably and also again back to that Loop picture of the entangle cycle uh is there just no passage of time no expenditure of energy is does that any of that make sense uh yeah and I think it's worth actually going to the next slide here where we complete the loop and add the quantum is it is it this one yeah I think to to understand what's going on here I think it's very useful to think of it in terms of uh of the lepr experiment right where Bob in this case the shared environment includes the source that's creating the entangled pair which of A1 and A2 observe they interact simultaneously with their two detectors which are interacting with the entangled pairs so you can think of their their two detectors are entangled and it's only later that they exchanged this classical information and in a bell APR experiment the the time the elapsed time between the source emitting the entangled pair and the entangled pair interacting with the detectors deployed by A1 and A2 is is designed as part of the experimental protocol to be less than the time that it is required for A1 and A2 to communicate classically and notice that from the point

of view of the entangled pair itself no time has passed at all right it's the integral pair is from a special relativity point of view moving along a light light trajectory so there's there's no distance being traversed in no time is is elapsing from the point of view of the entangled pair uh there's only an elapsed time and a Traverse distance in the coordinates of The Observers and it's those coordinates that Define the parameters for the observer's eventual classical communication so that's this is why I characterize the the Assumption of classical communication as a stipulation and also why I characterized it as a fine-tuning assumption right we're we're assuming something very specific about this communication Channel uh and we're assuming that the part of Bob the part of the shared environment that implements this classical Communication channel behaves differently from the rest of the shared environment uh it behaves in a way that is is classical uh but the rest of the shared environment doesn't right the rest of the world is is described using quantum theory so we're we're making this assumption that part of the shared environment is special and the the specialness is that it it behaves in this classical way so let me let me finish with this uh slide I suppose uh to go back to my earlier question about that entangled Loop being at least rhetorically appears to be like a degenerate version of this log picture if in this loud picture I have the quantum channel on the right uh that hypothetical entanglement on the to complete the cycle on the left was only broken when A1 and A2 chose to write that into their memories at different sectors right and then that eventually necessitated the classical communication between them is that is that correct yeah you know one could one could always take the point of view uh if you will The Observer who's outside the entire universe and just sees the entire universe evolving unitarily and in from that point of view there's no such thing as classical communication so the communication is classical only only relative to this division between The Observers and that division between The Observers corresponds to their their measurement procedures not commuting so it it corresponds to what we think of as Alice acting on her half of the entangled state uh determining what Bob will then see again that's that all of that causal language is classical uh and we're like at 11 30 so halfway more or less let's say a few words on uh the later slide but still uh it's getting profits and all of that I guess before we before we start talking about this which will take us like five to ten minutes at most would you say if you if you worked again about the fine tuning assumption so uh I still can't quite understand what is meant here that this whole classical communication staff requires fine-tuning okay I I think uh a way to think about it to to think about what it means to say that Alice and Bob are able to communicate what we're saying is that Alice and Bob share some sort of language so when

they exchange their data we can think of their data as as written in some binary code in a data table you know Alice says I set my instrument this way and I saw this and I then set my instrument this other way and I made this observation so Alice's data table or Alice's data table is just a long table with two entries in each row one entry is how the detector was adjusted and the other entry is what the value of the spin was observed and Bob's data table looks the same and so they share this information or they share it with some third party they communicate it classically and there are two aspects to that classical communication one is that up to classical noise of their data tables don't change when they share them so uh they really are irreversibly recorded they're not evolving in time uh along with the rest of the universe the other aspect is that we have to assume that Alice and Bob mean the same by their notations in these data tables so that they can actually compare them so that they can actually do the statistics and see if they've violated Bell's inequality so what that means is we have to assume that they actually share a definition of up and that when they talk about changing the settings on their instruments they're talking about changing those settings with respect to some shared definition of up and you know in a real experiment that shared definition of up is provided by the Earth's gravitational field or something like that Earth's magnetic field some shared environmental variable so now it seems like oh we have a situation in which Alice 1 and Alice 2 should be entangled right because they share this reference frame that enables them to communicate but they can't be entangled if they're going to communicate classically so we make an assumption that in just this case they share a reference frame but they're not entangled they're able to communicate classically while sharing the reference frame we can sort of wave our hands and say uh they only share the reference frame up to some measurement resolution or something like that despite the hand waving we're making a very specific assumption that violates the fact that reference frame sharing induces entanglement so that's the fine-tuning assumption we assume that the universe is such that in this very specific kind of case uh we can have reference frame sharing without entanglement cool let's keep it here and just open it up a little bit if um Harris or Dean want to just get any thought or reflection on what we've gotten to so far Paris if you want to please yes everything is very interesting um basically I was listening uh with great let's say focus it reminded me this let's say elaboration on the previous talk that Chris gave two weeks ago it reminded me something that Michael Levin said some time ago it is regarding this amazing ability of uh cognition to scale up so it's not only this understanding let's say of the scattering of cognition but also the scaling up of cognition and when I try to get inside in my

head actually that uh asked Chris said in in one of his talks that if a system is a toggle one cannot throw a meaningful boundary across it so these fine fine tuning let's say elaboration that was made right now actually helps me understand that even better so thank you Chris for that and thank you under for this elaboration well thank you thank you I'm glad that helped yeah I I think this is intimately related to the question of what cognition is since if if we have a if we have a purely Quantum system there there really is a deep sense in which nothing is happening because of the time symmetry so to talk about cognition is something that actually occurs uh we have to think of it classically thank you also if you allow me another comment on that and basically one of the comments posted during your last talk in the chat had to do with classical memories and I was thinking about that and I I came to the conclusion that classical memories presupposes a gestation period and and when I think a lot about it that actually makes sense and it is true let's say uh for for both physical and psychological reasons so uh for example a hard disk needs to write data on its magnetic surface and a human brain needs to consolidate all those neural connections through synaptic plasticity let's say however when we start thinking about quantity Quantum information and Quantum Quantum memories let's say we realize that such memories will be such Quantum memories basically will be stored in qubits or in those and toggle states that you described so the Indian necessarily have the same limitations and of course Quantum information theories suggests as I understand it that Quantum memories can be created and accessed instantaneously without any form of gestation let's say without any gestation period and of course this goes back and it has the potential as I see to revolutionize computation with in terms of quantum Community computation but also and most importantly communication and and Quantum cryptography even and this is something that I I am truly interested so yeah I just wanted to share also that yeah that's that's a very good point in the the flip side is the the issue of stability which again gets to the notion of of Environmental decoherence which is just the process of observation of the quantum state by uh some environment or some Observer some component of its environment and uh the reason this is a problem in practice is that we don't have complete control of all components of the we have to use things like liquid helium to try to keep the environment at Bay to prevent decoherence uh because any other components of the environment interacting with our Quantum memory will alter the state of the memory yeah that is uh that is a great and great answer and basically the reason why I type that question regarding super selection rules and namely I propose something called fuzzy super selection rules basically as I understand it those super selection rules the classical let's once let's say

they are based on symmetries or on on all those conservation laws that actually limit the accessibility over there our capacity to to mention let's say certain Quantum properties let's say maybe it's parity maybe it's angular momentum maybe it's even charged but if we if we start thinking like this fuzzy super selection rules in an effort to relax these restrictions by allowing some degree of coherence or interference between the different super selected sectors which can be a Quantified let's say by a parameter called fuzziness I know that in other things like in my field that is fuzzy Cooperative mapping the production of fasiness from us as it was propounded by Professor ludwizadev it actually helped a lot of our models to become a more accurate let's say so this is what I am proposing maybe of course I I don't expect it to revolutionize Quantum information but maybe it can help us understand a bit better what is going on in the role of this the coherence that comes out of the environment so if we incorporate it as a parameter of fuzziness a priori then maybe we can understand and even what we are dealing with uh yeah I I would refer you to the paper by Bartlett um who are the other there are two other co-authors in 2007 uh a lovely review of quantum reference frames in reviews of modern physics and a point that they make that I think is relevant here is that a super selection rule can always be interpreted let me see if I if I I'm not sure I'll be able to completely reproduce their argument but they showed that a super selection rule can always be interpreted as the absence of a reference frame or the the lack of a reference frame and one of the examples that they give um charge conservation that we never see superpositions of different electric charges and uh their point is we never see this because we have no way of measuring it we have no reference frame that we could use that would allow us to see a state of of superposed charges and effectively what they're saying is our reference frames are all made of matter that uh is is that is composed of stuff that has well-defined electric charges so we can't we can't see things that are superpositions of charge and it's very difficult you know for us to see things that are superpositions if positioned this question of what is a super selection rule why do we have these rules at least in their reasoning boils down to the question of why do we have some reference frames or sun measurement capabilities and not others what is it about us limits in some fundamental way our ability to make measurements I suspect that when you talk about buzziness which is a sense of of classical uncertainty that you may be sort of pointing in that direction but I think I think it would be worth looking at that paper and which is which is largely written in English right the the formalism is oh wow so it's a it's a very nice paper hey thank you once again thank you um Andrew maybe mute a little bit Dean on the Mountaintop what can you throw down on us everything super special throw down I I

have a I have a question though and because I am very comfortable in that fuzzy space I kind of see relationships uh and their and their lack of definability as a feature so here's my here's kind of my question so you talked about a minimum sort of two different memories to encode something in time and I think it's it was one of the slides just after the one that's up here right now which sounds to me like somewhere processing capacity it sounds like we are kind of stepping back and trying to look at this instantaneous from the relative position of even the things that are classical can be seen as all Quantum so that when you mentioned that earlier Chris that helped me quite a bit which then manifests itself as a classical ability to process somewhere I I spent a lot of my career self-labeling as a way finder so not capacity realization relationship just as a relationship it means that wave binding is not just an orienting or a processing where I am and being gripped by the sort of the things I'm enveloped by it's also the channel the screen the bound the divide the Gap the unentangle potentiator it's the thing upon which traction can exist as being a gripper as well so if it's I was already hung up on I wanted to go back like Colombo in that old 70s TV show and talk about z-spin because I think z-spin is an energy way of describing but I honestly think when I went back and watched the your presentation a couple of times I'm starting to get a sense that z-spin is also a relationship it's that affordance that allows us to be able to see up and down but I'm just curious Chris are we really talking about processing somewhere in the classical sense but because we have these two ways of encoding different memories we're also talking about somewhere processing just that state that allows for that that's my my question is that what we're really getting are we are we trying to open that up that possibility uh yeah I I think that's a nice way to think about it I think we're trying to at some deep level come up with a way of describing what it means to be an agent uh and I'll say in our our case a self-conscious agent uh with a sense of time that's embedded in an environment that we depend on to store memories for us and what I what I mean by that is that we we depend on the environment change very slowly at least from our point of view so that we can keep some sense of our orientation within the environment right we're using everything we see as a memory that allows us to keep our ourselves fixed which allows us to have a point of view so we're we're using the entire visible environment as if you will a reference uh against which we can see small changes but if the environment was constantly changed constantly and rapidly changing we wouldn't be able to do that and we wouldn't be able to use it as a memory and so we would have no point of view so in a sense I think what we're talking about here is the kind of of the the miracle of having a point of view that gives us a sense of time that's very philosophical that it's it's amazing and I wonder if

along with the perspectival visual advancements which you and Shannon Dobson have explored in the math art settings you're almost bringing that kind of a transformative re-understanding or re-quantum reference framing into a temporal setting where it's not just a static spatial parallax that we are aesthetically and formally re-understanding but also a temporal one which in an agent environment system necessitates these questions of which in the quantum information formalism are approached topologically rather than geometrically which is what we needed for the kind of vanishing point type math art in the spatial setting well I think their approach geometrically eventually and and that's really what the session in September will be about it's how to start with this very topological picture where we're only talking about interactions in a sense interactions and time but but outside of space and layer of spatial structure on top of I I think this is particularly interesting in this active inference context because it immediately raises biological questions about um the the ability of different organisms to layer space on their experience and hence locate themselves if you will within some sort of perceived world awesome Dean and then back to under the last minutes and then please continue sorry so Chris is it from the pure neophyte perspective that I have is it okay to say that it's not just a flipping of position of words that somewhere processing that availability and processing somewhere in the classical sense where am I it's the it's our ability to fuse those as you mentioned we can we can slow down we can stabilize we can see that relationship or is it about spin I think the former right spin is a very convenient abstraction for talking about binary encoding and we have this felt sense of spin but the sense it's use in physics is very much an abstraction I mean the difference between a proton and a neutron is a spin variable awesome under you can in last minutes take it where you need also on mute and then continue can you guys hear me now well so let's uh hope that I have a time for Chris to address a couple things so just to conclude the slides um and then hopefully we get to say a few words on Quantum Darwinism so the towards the end the slides um appear to start pivoting towards the next subject which is space and time and here in particular we see the earlier slide uh showcasing the log protocol where you can you can notice that half of it has been cut and he's has been moved to the Future so here we don't have uh two sectors of powerless communicating but rather Alice communicating with its future still you have the classical and the quantum Channel now one question that I brought up I remember and uh in the lecture and maybe we can expand on a little more was exactly what is the difference between this Quantum and the classical Channel part um something I heard Chris say is that it is crucial for the quantum channel to be coherent so this is going to be some sort

of unitary scattering Evolution that is completely independent of everything else going on uh is that correct or should be in theory and ask yes yeah and and I suppose just like earlier the classical the classical channel is is the part that takes time and free energy expenditure now uh my question uh to you Chris is on no no not the not the channels in the middle but the ticks on either side of the boundaries that have been broken down and separated in terms of an external time what's what's up with those sticks from one sector to the other of the binary so in q1 for example the allow the lower CRF sector that's that amount to a state was prepared the upper one was I made a memory record of that state is that that's what it is right okay so uh in in the lower sector we both had a preparation and a measurements and a measurement was recorded classically that's all there is yeah so so preparation and measurement are are effectively the same process um and you can think of uh in A1 of t what A1 is doing from an operational point of view she's preparing an instrument and turning it on she's she's creating an initial state and she's simultaneously encoding a memory someplace in her head or or on a piece of paper of what does Preparation conditions were because it's only with respect to the memory of what the preparation conditions were that the later measurement is Meaningful to let's let's go back to just preparing a spin uh right in at T she may uh set the angle of some polarization filter that defines a propagating beam of photons and at t plus delta T she measures something about what those photons are doing but the measurement is only meaningful with respect to her memory of what the polarization setting that prepared the photons was so the effectively what's going on in the quantum channel is an experiment and it's it's the the classical information that's preserved in this classical channel that makes the experiment meaningful or interpretable or informative yeah for sure yeah that makes perfect sense and and preparation of measurement this dies back to an earlier question we had what would happen simultaneously at the same um uh degree of Freedom At The Boundary or do we have preparation at the lower bounds are the top of one the only reason you so yeah when one can one can think of that operator M_i if you remember the decomposition of the hamiltonian as as preparing or measuring that it's just a matter of interpretation uh you know effectively is it acting to the left or acting to the right yeah okay yeah yeah yeah and then um before we get to Quantum American Specialists instead of a time thing I wanted to ask God you have another question between this one so the quantum channels does involve some sort of time Evolution you know it's this unitary you know portion that is uh sorry this portion that is evolving unitarily with no interference from the outside uh however as you said for this experiment to make sense you need to have a classical memory of how that experiment most prepared plus that means that

that classical guy on top is uh aesthetic there's no time flowing it uh well this actually gets back to the talk by suskin that we were talking about before we started here uh that you you kindly referred me to part of part of this classical memory is uh memory for what counts as a classical clock and we'll we'll actually get to this again in the September session but if you if you think again about doing an experiment uh in which a state is prepared and then it's measured uh this Quantum process is a process Through Time with respect to some uh classical clock that actually provides the time coordinate for the experiment this classical clock is some external reference frame that A1 here in this notation is using as part of her classical memory right again uh it's a Memory because it's something that's assumed not to change and and this is this will be kind of the heart of the discussion in September that that time and memory and object persistence are all different names for this same assumption that we make about the world mm-hmm thank you Chris thank you I I am going to have to disappear so only for now though yeah only for now I I will I will nonetheless persist someplace come on Chris okay awesome thanks very much well measure you later okay um enter if you wanted to drop any other cues or clues or bring us a little bit of a preparation for where we're going to be heading in lecture five that would be pretty cool uh sure so I mean I I'm a little aware of what's coming in lecture five until about uh space sometime on how space may be sorry time may be a little more fundamental than space for reasons that Chris will show has a lot to do with the notion of how you define a clock as in we need to think hard about what what what are the bare minimum ingredients for you to talk about the existence of a clock uh these are deep fundamental almost philosophical questions that are overlooked and the good news and that's the sasking talk that Chris and I were referring to referring to is that other people especially in the high energy Community are also looking into these questions which is very reassuring so I'm talking uh Harlow Etc they are there's a recent talk on YouTube where they talk about what does it mean for something to be a clot you know how that's how long does a clock last and and then Chris will also talk about space and and the punchline of it will be there's obviously a physics approach to emerging space-time which is very very technical and and so on but from what I understood I mean I I have yet to see the talk my first my thoughts myself first um is that there is a parallel biological approach as to how we learn our space time so from uh developmental psychology point of view influence and I suppose all kinds of animals are not born uh fully equipped to perceive space-time hence the necessity of things like clay and so on so they can find correlations between uh motor inputs and visual inputs right and eventually things like the mirror test which is non-trivial no

no no no no no human born is born uh passing the mirror test already so that implies that at least biologically it is non-trivial and it takes a few steps until you get to have that sort of a space spatial awareness right now the interesting part of this for me personally is with the with the physics side of it is and uh I I'm no expert by any means but I think the gist of it might might have to do something with Quantum error correction uh ads CFT and so on um there was a bit of a teaserral at quantumeric correction here in this slides uh but yeah I mean that's that's about all I can say but very looking forward to the biology also yeah cool Harris any last thoughts or anything you want to point towards basically I would like to thank everyone for inviting me I really enjoyed it the whole uh live stream uh I hope we have made some progress and looking forward to the next live streams number five and number six so that's about it thank you everyone thank you anything else Andrew that's it awesome well I encourage everyone to go to the syllabus Page look ahead look back pre-gest digest reject ask questions and let's make progress as you exhort Paris so see y'all until next time